

Math 514, F13
Quiz 7

Name: Answer
SSN: xxx-xx-

Instructions: You must show all of your work to receive a credit for a correct answer.

Problem 1. (6 points) A ten-year \$10,000 bond costs \$10,000 and pays its holders \$500 every six months for ten years with a final additional payment of \$10,000 made at the end of those twenty payments. Assume the interest rate is 6% and is compounded continuously. Find the present value of the bond.

The effective interest rate per six months is

$$r_e = e^{\frac{0.06}{2}} - 1 = 0.03045$$

Thus

$$\begin{aligned} PV &= -10,000 + \sum_{i=1}^{20} \frac{500}{(1+r_e)^i} + \frac{10,000}{(1+r_e)^{20}} \\ &= -10,000 + 500 \cdot \frac{1 - \frac{1}{(1+r_e)^{20}}}{r_e} + \frac{10,000}{(1+r_e)^{20}} \\ &\approx 2895.69 \end{aligned}$$

Problem 2. (4 points) Find the rate of return for an investment that for an initial payment of 100 return 50 at the end of 1 year and additional 60 at the end of 2 years.

$$P(r) = \frac{50}{1+r} + \frac{60}{(1+r)^2} - 100 = 0$$

let $\beta = \frac{1}{1+r}$ we get

$$60\beta^2 + 50\beta - 100 = 0$$

$$\Rightarrow \beta = \frac{-50 \pm \sqrt{50^2 - 4 \cdot 60 \cdot (-100)}}{120} \approx 0.9399$$

$$\Rightarrow r = \frac{1}{\beta} - 1 \approx 0.06394$$